

Family Tabanidae

Horse flies

Occurrence 4,000 spp. worldwide, near mammals; many favor damp or wet habitats

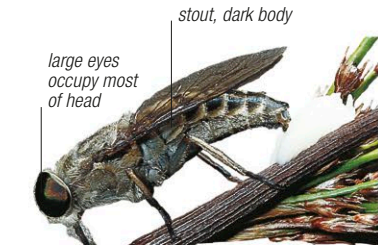


Horse flies are stout and hairless with a distinctive, flattened, hemispherical head and colorful, patterned eyes. Most are black, gray, or brown, with brightly marked abdomens, and measure ¼–1¼ in (0.6–2.8 cm) in length. Females have bladelike mouthparts for cutting skin; they feed on the blood of mammals and birds. Eggs are typically laid in a



TABANUS SUDETICUS
This giant horse fly has a beelike abdomen with distinctive, pale, triangular marks down the middle. Some species transmit diseases to animals as well as humans.

mass of 200–1,000 on debris overhanging water, leaves, or rocks. Larvae are predatory or saprophagous.



TABANUS SP.
These dark and robust flies often mimic large bees, such as the carpenter bee, in order to deter predators.



CHRYSOPS RELICTUS
This twin-lobed deerfly inhabits bogs and ponds in heather moorland in the Northern Hemisphere. Its bite is very painful.

Family Tephritidae

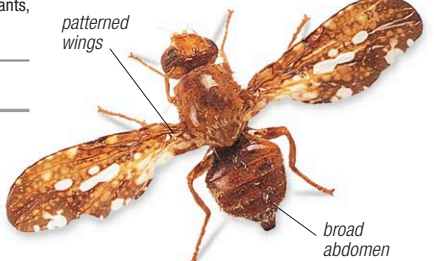
Fruit flies

Occurrence 4,500 spp. worldwide; on a range of plants, crops, rotting matter

Habitat Terrestrial

Most fruit flies are under ½ in (1.5 cm) long and recognizable by their distinctive wing patterns, which can take the form of bands, patches, and zigzag markings. Females have a pointed ovipositor, which can be longer than the rest of the body. Males display to females by waving their wings. Eggs are laid singly or in groups

under the surface of fruit. Adults take fluids, plant sap, nectar, or liquid from rotting matter; the larvae are herbivorous. Many species are crop pests.



ICTERICA WESTERMANNI
Found in hedgerows, waste ground, and grassy places in Europe, this species of fruit fly feeds on the flowerheads of ragworts.

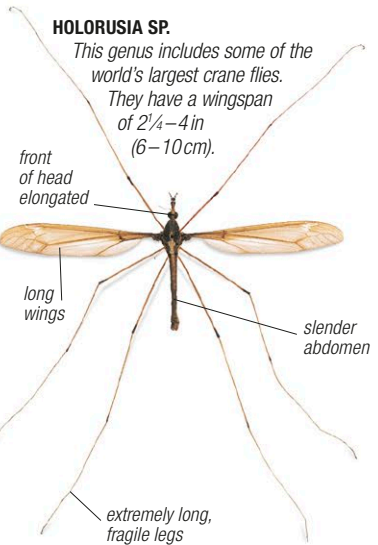
Family Tipulidae

Crane flies

Occurrence 15,000 spp. worldwide; usually in moist microhabitats

Habitat Terrestrial

Also known as daddy long legs, most of these fragile, slender, and elongated flies measure ¼–2¼ in (0.6–6 cm) in length. They shed their legs very easily if caught. The adults of many are short lived and feed on nectar and other fluids. The tough-bodied larvae, called leatherjackets, live in soil, rotting wood, bird nests, and bogs, and are a food source for many animals and birds.



HOLORUSIA SP.
This genus includes some of the world's largest crane flies. They have a wingspan of 2¼–4 in (6–10 cm).

Family Tachinidae

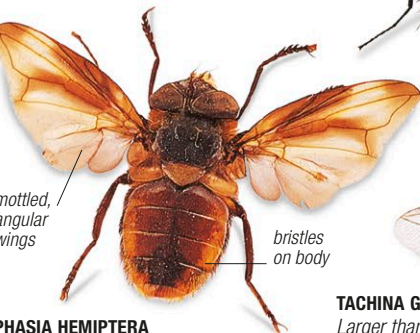
Parasitic flies

Occurrence 8,000 spp. worldwide; where insect hosts are found



Stout and robust, parasitic flies are ¾–¾ in (0.5–2 cm) long, and are very variable in appearance. Most are dark; some are metallic or pale. The abdomen is very bristly, especially toward the rear. Many species look like bristly house flies (see muscids p.567) and some species are almost beelike. Adults feed on sugary fluids,

but their larvae are endoparasites of insects. Some species locate crickets to parasitize by listening for their mating songs. Males and females often congregate on hilltops for mating, and the eggs are laid on foliage eaten by hosts, or inside the hosts' bodies. Parasitic flies are mostly diurnal. Many species are used as biological control agents of herbivorous insect pests.



PHASIA HEMIPTERA
An inhabitant of meadows and woodland in parts of Europe, this species parasitizes shield bugs.



ERIOTHRIX RUFOMACULATA
A native of Europe and Central Asia, this species parasitizes caterpillars of tiger and ermine moths, as well as lappet and eggar moths.

TACHINA GROSSA
Larger than most of its relatives, this fly measures ¾ in (1.9 cm) in length. It is found in Europe and Asia.

Order Trichoptera

Caddisflies

There are about 10,000 species of caddisflies, in 43 families, and they can be found almost anywhere there is freshwater. The adults are slender, dull-colored insects, with a hairy body and wings, and long antennae. The larvae are soft-bodied, and live in water, often inside a protective, portable case, which each species builds in a characteristic way. They develop by complete metamorphosis.

Family Limnephilidae

Northern caddisflies

Occurrence 1,500 spp. mainly in the Northern Hemisphere; around lakes, temporary pools, ditches



Northern caddisflies are reddish, yellowish, or dark brown, with dark wing markings, and most are under 1 in (2.4 cm) long. Larvae eat organic detritus and algae, and make cases from sand, pebbles, vegetation, or snail shells. Adults may feed on liquids.



LIMNEPHILUS LUNATUS
A widespread species, its larvae are found in a wide variety of freshwater habitats. Coloration ranges between drab shades of black and brown.

Family Phryganeidae

Large caddisflies

Occurrence 450 spp. mainly in the Northern Hemisphere; near ponds, lakes, bogs, slow-moving streams, rivers



Members of this family have light brown or gray markings, and may look mottled. They are ¾–1 in (1.2–2.6 cm) long and have at least 2 tibial spurs on the forelegs and 4 on the mid- and hindlegs. Larvae make cases out of plant fragments and fibers.



PHRYGANEA GRANDIS
This is the largest caddisfly found in the UK. The male, smaller than the female (shown here), lacks the distinctive dark stripe on the wing.